# Thu Htet

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## EDUCATION

**University of California, Davis | GPA 3.45** **Davis, CA**
**Bachelor Of Science – Electrical & Computer Engineering** **Expected March 2027**

- Relevant courses: Circuits, Embedded Systems, Testing and Verification of Digital Systems, Digital Systems, Computer Architecture, Computer Networks, Algorithm Analysis

## EXPERIENCE

**Space and Satellite Systems — Battery Management System** **December 2024 - November 2025**

- Designed a 4S lithium-ion battery management system featuring battery protection, cell monitoring, and power management functions for CubeSat applications
- Created schematics and PCB layouts in KiCad, integrating protection ICs, sensing circuitry, and power distribution networks while following PCB design best practices
- Built and validated hardware prototypes through soldering, debugging, and environmental testing, including thermal, short-circuit, and vibration evaluations.

**Assista Robotics — Undergraduate Robotics Researcher** **January 2025 - July 2025**

- Presented the robotic platform at the 2025 UC Davis ECE Picnic Day Showcase, earning 1st Place.
- Developed embedded firmware for a 9-axis robotic arm platform, implementing control logic and system integration for autonomous fryer and grill automation in commercial kitchen environments.
- Integrated embedded software with sensors, actuators, and robotic subsystems while debugging, and refining system performance through iterative testing.

**De Anza Computer Science Department — Teaching Assistant** **January 2023 - June 2024**

- Assisted over 100 students across Java, C++, Data Structures, Algorithms, and Object Oriented Programming courses during labs and office hours.
- Debugged programming assignments, explained software engineering concepts, and helped students improve problem-solving and coding practices.
- Worked closely with instructors to support grading, answer technical questions, and create an effective learning environment.

## PROJECTS

**Smart Home Automation System - PCB Design & Embedded Systems** **June 2026**

- Designed a modular smart home control interface featuring buttons, rotary encoders, and magnetic expansion modules for configurable device control
- Developed a compact PCB integrating a microcontroller, user input components, and external interfaces to support flexible expansion with additional modules
- Implemented I2C-based communication between microcontroller and peripheral modules for real-time control and status monitoring.
- Received UC Davis ECE Outstanding Senior Design Project Award

**Smart Fridge Device — Embedded Firmware & Circuit Testing** **February 2026**

- Developed embedded C firmware on a PSoC microcontroller to monitor refrigerator door status and temperature, generating real-time visual and audio alerts
- Integrated thermal and magnetic sensors with ADC and GPIO peripherals to enable real-time environmental monitoring and event detection
- Configured and debugged I2C, GPIO, timers, and ADC peripherals in PSoC Creator, validating hardware–software interaction on prototype hardware

## SKILLS

- **Programming: & Embedded Systems:** C, C++, Python, Java, Embedded C, ESP32, STM32, PSoC, FreeRTOS, I2C, SPI, UART, GPIO, ADC, Timers, Interrupts
- **Hardware/PCB:** KiCad, Altium, Cadence Virtuoso, LTSpice, PCB Layout, Schematic Design, Soldering, Oscilloscope, Multimeter, Logic Analyzer